GGG-S-1808
April 23, 1973
SUPERSEDING
Fed. Spec. GGG-S-121D (In Part)
March 3, 1959
(See 6.2)

FEDERAL SPECIFICATION

SCREWDRIVER, JEWELERS'

(SWIVEL KNOB, INTERCHANGEABLE BLADES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers flat tip screwdrivers of the type normally used by watch makers, opticians, and instrument makers.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and sizes. Screwdrivers shall be of the following type, and of the sizes as specified (see table I and 6.1):

Type I - Swivel knob, interchangeable blades.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Services Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 5120

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI), Inc., Standard:

B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of screwdrivers which are otherwise in accordance with the requirements of this specification.

3.2 Blades. Blades shall be of high-grade steel and properly heat-treated to withstand the tests specified herein. The cross-sectional shape of the blade shall be round. The blades shall be free from visible defects such as scale, seams, laps, and cracks.

3.2.1 Finish. The blades shall have a smooth, bright polish finish that will produce a 63 microinch roughness height rating, or better (as determined in accordance with B46.1), on all exposed portions other than the tip vicinity. The blades shall be treated in a manner to prevent rust or corrosion. Plating, if any, shall be adherent. There shall be no evidence of plating peeling, or chipping. Cadmium plating will not be acceptable.

3.2.2 Blade hardness. Each blade shall show a hardness of not less than 48 nor more than 58 Rockwell C. The reading shall be taken as near the tip end as practicable. The blade shall be checked for hardness by file method (see 4.4.1).

3.3 Type I, swivel knob, interchangeable blades. Each screwdriver shall consist essentially of a body, chuck jaw, binder nut, swivel knob, and a blade. The design shall permit withdrawal of the blade by loosening the binder nut.

3.3.1 Body. The body of type I screwdrivers shall be of steel or aluminum, cylindrical in shape, and neatly knurled to provide a comfortable handgrip. One end of the body shall be drilled to receive the chuck jaw and the other end shall have a swivel-type knob. All parts subject to corrosion shall be protected by nickel plating or other suitable and equally effective means, except cadmium plating will not be acceptable.

3.3.2 Chuck jaw. The chuck jaw shall be made of steel, with at least two slots for jaw flexibility. The slots shall extend at least one-half the length of the chuck jaw. The outside of the jaw shall be threaded to engage the binder nut properly and hold the blade securely. The chuck jaw shall be securely fastened into, or be integral with, the body.

3.3.3 Binder nut. The binder nut shall be of steel and threaded internally to securely fit the chuck jaw. The major diameter of the nut shall be neatly knurled for a minimum length of 3/16 inch. The remainder of the nut shall be smoothly and uniformly tapered.

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3.3.4 Swivel knob. The swivel knob shall be hexagonal or square in shape. It shall be of metal or plastic such as phenolic, allyl, polyester, or polyamide resin. It shall be securely attached to the body, but left free to rotate. The knob shall be not less than 1/2 inch across flats.

3.3.5 Blade. The blade shall be in accordance with 3.2, 3.2.1, and 3.2.2. The blade tip(s) shall be of the type known commercially as cabinet tip, the straight sides being approximately parallel to the blade shank. The extreme shank end of each blade shall be flattened on two sides to seat firmly and shall be a good fit in the chuck jaw. The tip end shall be at right angles to the longitudinal axis of the blade within 3°, and the tip thickness shall be uniform across its entire width within 0.003 inch. The included angle between the bearing surfaces on the tip end shall not exceed 20° in order to minimize cam-out action from screw slots. Blade hardness shall be as specified in 3.2.2.

3.3.6 Type I screwdrivers shall conform to the dimensions of table I for the width of tip specified (see 6.1), and be similar to figure 1.

TABLE I. Type I screwdriver, jewelers', swivel knob, interchangeable blades

Blade				Handle	
Width of tip ±0.005	Thickness of tip at end +0.003 -.002	Diameter at chuck end ±0.002	Length under chuck +1/2 -1/4	Diameter ±1/64	Length +1/4 -1/8
Inch	Inch	Inch	Inches	Inch	Inches
0.025	0.004	0.100	7/8	1/4	3-1/2
.040	.004	.100	7/8	1/4	3-1/2
.055	.006	.100	7/8	1/4	3-1/2
.070	.006	.100	7/8	1/4	3-1/2
.080	.008	.100	7/8	1/4	3-1/2
.100	.010	.100	7/8	1/4	3-1/2

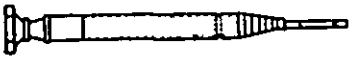


FIGURE 1. Type I screwdriver, jewelers', swivel knob, interchangeable blades.

3.4 Sets. When sets of type I screwdrivers are required, they shall be furnished in the sizes as specified (see 6.1). Unless otherwise specified (see 6.1), a set shall consist of six handles and six blades. Sets shall be furnished in a plastic wrap, roll, tube, or fitted into a box. When a wrap or roll is provided, it shall have separate pockets for individual screwdrivers or blades.

3.5 Individual blades or handles. Individual blades or handles shall be furnished in the size(s) required as specified in 6.1.

3.6 Identification marking. Each screwdriver shall be marked in a legible and permanent manner with the manufacturer's name, or with a U. S. registered trademark.

3.7 Workmanship. Screwdrivers shall be symmetrically made, smoothly and properly finished, and free from any defects which may affect their serviceability, durability, or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

TABLE II. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	1 ea.	II	2.5	Defects per hundred units	4.3.1
Dimensional examination	1 ea.	S-3	2.5	Defects per hundred units	4.3.2
Testing	1 ea.	S-3	2.5	Percent defective	4.4

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 Hardness. The hardness of the blades shall be determined by the use of two test files having a Rockwell C hardness of 48/49 and 57/58, or such test files as are available. The blade to be tested shall be rigidly mounted in a suitable clamping device and the test file drawn across the blade. The extent of cutting or abrading of the surface of the blade shall be compared with similar markings on a standard blade of such size that the Rockwell C hardness is readily obtainable for comparison. The file shall be in good condition and shall be applied to the surface with short, firm strokes attempting to "feel" how the file bites the surface. Hardness of the blade should be such that it cannot be cut or abraded with the soft file, but can be cut or abraded with the hard file.

4.4.2 Surface finish test. Each of the sample screwdrivers shall be tested for surface finish in accordance with ANSI B46.1 to determine compliance with the surface finish requirements specified in 3.2.1.

4.5 Inspection of preparation for delivery. Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with PPP-P-40.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Sizes required (see table I).
(c) Sizes required for sets (see 3.4).
(d) Specify if other than six handles and six blades are required (see 3.4).
(e) Individual blades or handles as required (see 3.5).
(f) Specify level of preservation, packaging, and packing (see 5.1).

6.2 Supersession data. The type of screwdrivers covered in this specification corresponding to those covered in Federal Specification GGG-S-121d is as follows:

This specification	GGG-S-121d
Type I - Swivel knob, interchangeable blades.	Type II - Jewelers'.
Deleted.	Class 1 - Swivel knob, interchangeable blades.
Deleted.	Class 2 - Nonmetallic handle, nonremovable blade.

MILITARY CUSTODIANS:	User activity:	Preparing activity:
DOD Waiver Coordination	Army - GL	GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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